Wantida Horpiencharoen, PhD, DVM

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Tertiary Education

PhD in Veterinary Epidemiology **2020 - 2024**
School of Veterinary Sciences, Massey University, New Zealand
Dissertation: *Wild bovid habitat and infectious disease risk in Thailand*

MS in Veterinary Epidemiology **2016 – 2018**
Faculty of Veterinary Medicine, Kasetsart University, Thailand
Thesis: *Bayesian network and logistic regression analysis on clinical mastitis in dairy cows raised by small-holder farmers in western part of Thailand*

Doctor of Veterinary Medicine (DVM) **2006 – 2012**
Faculty of Veterinary Medicine, Kasetsart University, Thailand
Research project: *Sex identification of Red-whiskered Bulbul (Pycnonotus jocosus) by PCR technique*

Work Experience

Postdoctoral Researcher **April 2024 - Present**
Institution: Department of Ecosystem Science and Management, The Pennsylvania State University, USA
Project: *Targeted Surveillance for SARS-CoV-2 in White-tailed Deer*
Responsibilities: Conduct epidemiological and GPS data analyses, develop deer density estimation models, and examine spatial ecology, movement patterns, and contact networks of white-tailed deer exposed to SARS-CoV-2 in Pennsylvania and across nine U.S. states.

Field Researcher **2019 - 2020**
Institution: The Golden Triangle Asian Elephant Foundation and Hunter College, City University of New York
Project: *Understanding Elephant Behaviour: A Novel Approach to Mitigating Human-Elephant Conflict in Salakpra Wildlife Sanctuary, Kanchanaburi, Thailand*
Responsibilities: Collect behavioural data on wild elephants inside and outside forest areas, install and monitor problem-solving puzzle boxes, and identify individual elephants. Support field operations by setting up and maintaining camera traps and contribute to project reporting through preliminary data cleaning and analysis.

Research Assistant **2018 – 2019**
Institution: Faculty of Veterinary Medicine, Kasetsart University, Thailand
Project: *Animals Free of Rabies: Humans Are Safe from the Disease*
Responsibilities: Conducted dog population surveys and implemented a catch–neuter–vaccinate–return (CNVR) programme for roaming dogs on campus. Managed field sampling and performed epidemiological analyses to inform rabies vaccination strategies and the design of dog population management and control programmes.

Veterinary Clinician **2012 - 2015**
Institution: Livestock and Wildlife Hospital, Faculty of Veterinary Science, Mahidol University, Thailand
Responsibilities: Provided routine clinical care, health management, disease diagnosis, and medical treatment for domestic livestock and wildlife. Conducted field and clinical investigations of disease outbreaks, applying veterinary expertise to improve herd hygiene and husbandry practices. Supported the wildlife rehabilitation unit, providing care for native and trafficked wildlife.

Teaching Experience

- **Teaching assistant** **2018**
  Course: Biostatistics in Veterinary Epidemiology, Veterinary student course
  Institution: Faculty of Veterinary Medicine, Kasetsart University, Thailand

Funding and Awards

- **Funding** - Manaaki New Zealand Scholarships (2020 - present) $187,794 NZD - Ministry of Foreign Affairs and Trade (MFAT), New Zealand - Fully funded PhD scholarship

  Advisor: David T. S. Hayman. Co-advisor: Jonathan Marshall, Renata Muylaert, Reju Sam John

- **Funding** - Massey University Doctoral Conference Grants (2023) $1500 NZD funding for the oral presentation in MODSIM 2023, Darwin, Australia

- **Funding** - School of Veterinary Science postgraduate student fund:
  - 2023 - Oral presentation in MODSIM 2023, Darwin, Australia ($3000 NZD)
  - 2022 - Oral presentation in KVAC 2022, Khon Khen, Thailand ($1000 NZD)
  - 2020 - Resources (Computing equipment) (NZ $500 NZD)

- **Funding** - The Sakura Science Program by Osaka Prefecture University, Japan - visiting and presenting the research activities in Japan from 4 to 10 March 2018 with the Faculty of Veterinary Medicine, Kasetsart University

- **Funding** - Kasetsart Veterinary Development Funds (2016 - 2017) by Faculty of Veterinary Medicine, Kasetsart University - supported my Master's project - $4,000 NZD

- **Award** - The 23rd Khon Kaen Veterinary Annual International Conference (KVAC) - Best oral presentation Award (2022): Identifying remaining habitats suitable for wild bovids in Thailand - $150 NZD

- **Award** - The 20th case-study presentation, Faculty of Veterinary Medicine, Kasetsart University - Best oral presentation Award (2011): Sex identification of Red-whiskered Bulbul (*Pycnonotus jocosus*) by PCR technique

Publications

Google Scholar | **Orcid**

Published

- Giglio, R.M., Westmoreland, A., Wilber, M., Wilson-Henjum, G., Chan, A.N., Gardner, B., **Horpiencharoen, W**., Gagne, R.B., Corondi, A., Baker, A. and Combs, M., 2025. Viral outbreak dynamics and evolution in wildlife at the interface with humans. *Biology Letters*. https://doi.org/10.1098/rsbl.2025.0540

- **Horpiencharoen, W**., Marshall, J. C., Muylaert, R. L., John, R. S., & Hayman, D. T. S. (2025). Spatial risk of pathogen transmission from cattle to vulnerable and endangered wild bovids in Thailand. *Conservation Science and Practice*. https://doi.org/10.1111/csp2.70111

- **Horpiencharoen, W**., Muylaert, R. L., John, R. S., Marshall, J. C. Lynam, A., Ngoprasert, D., Godfrey, A., Riggio, A., Gale, G. A., Ash, E., Bisi, F., Cremonesi, G., Clements, G. R., Yindee, M., Shwe, N. M., Pin, C., Gray, T. N. E., Aung, S. S., Nakbun, S., ... & Hayman, D. T. S. (2024). Mapping threatened Thai bovids provides opportunities for improved conservation outcomes in Asia. *Royal Society Open Science*. https://doi.org/10.1098/rsos.240574

- **Horpiencharoen, W**., Marshall, J. C., Muylaert, R. L., John, R. S., & Hayman, D. T. S. (2024). Impact of Infectious Diseases on Wild Bovidae Populations in Thailand: Insights from Population Modelling and Disease Dynamics. *Journal of The Royal Society Interface*. https://doi.org/10.1098/rsif.2024.0278

- Jacobson, S. L., Dechanupong, J., **Horpiencharoen, W**., Yindee, M., & Plotnik, J. M. (2023). Innovating to solve a novel puzzle: Wild Asian elephants vary in their ability to problem solve. *Animal Behaviour*. https://doi.org/10.1016/j.anbehav.2023.08.019

- Yindee, M., Thomas, W., Wajjawalku, W., Wongsawang, W., Income, N., **Horpiencharoen, W**., ... & Ketpun, D. (2023). Clinical haematology, biochemistry profiles and erythrocytic morphometry of confiscated Sunda pangolins (*Manis javanica*) in the Centre for Rehabilitation and Quarantine of Confiscated Animals, Wildlife and Alien Species, Thailand. *Veterinary Integrative Sciences*, 21(2), 607-622. https://doi.org/10.12982/VIS.2023.043
- **Horpiencharoen, W**., Thongratsakul, S., & Poolkhet, C. (2020). Risk factors associated with environmental mastitis in clinical cases at small dairies in western Thailand. *Agriculture and Natural Resources*, 54(2), 190-196. https://li01.tci-thaijo.org/index.php/anres/article/view/242185
- **Horpiencharoen, W**., Thongratsakul, S., & Poolkhet, C. (2019). Risk factors of clinical mastitis and antimicrobial susceptibility test results of mastitis milk from dairy cattle in western Thailand: Bayesian network analysis. *Preventive Veterinary Medicine*, 164, 49-55. https://doi.org/10.1016/j.prevetmed.2019.01.014

Presentations

- **Horpiencharoen, W**., Giglio, R., Pepin, K. M., Wehr, N. H., Wilson-Henjum, G., Corondi, A., Walters, T. S., & Walter, W.D. (2025, April 21 - 24). *Contact network of white-tailed deer exposed to SARS-CoV-2 in a residential community in central Pennsylvania* [Oral presentation]. The 80th Annual Northeast Fish & Wildlife Conference, NH, United States. https://www.neafwa.org/schedule-2025.html
- **Horpiencharoen, W**., Marshall, J. C., Muylaert, R. L., John, R. S., & Hayman, D. T. S. (2024, October 19 – 23). *Assessing Wild bovid population dynamics and impact from infectious diseases in Thailand* [Oral presentation]. The Wildlife Society's 31st Annual Conference, Baltimore, MD, United States.
- **Horpiencharoen, W**., Marshall, J. C., Muylaert, R. L., John, R. S., & Hayman, D. T. S. (2023, 9 - 13 July). *Mapping the risk of disease transmission between wild bovids and domestic livestock in Thailand* [Oral presentation]. The 25th International Congress on Modelling and Simulation, Darwin, Northern Territory, Australia. https://doi.org/10.36334/modsim.2023.horpiencharoen
- **Horpiencharoen, W**., Marshall, J. C., Muylaert, R. L., John, R. S., Barraclough, R. K. & Hayman, D. T. S. (2022, September 1-2). Identifying remaining habitats suitable for wild bovids in Thailand [Oral presentation]. The 23rd Khon Kaen Veterinary Annual International Conference, Khon Kaen, Thailand.
- **Horpiencharoen, W**., Marshall, J. C., Muylaert, R. L., John, R. S., (2021, 7-8 December). *Habitat suitability and protected areas for endangered bovids in Asia* [Poster presentation]. The 7th One Health Aotearoa Symposium, New Zealand.
- **Horpiencharoen, W**., Dechanupong, J., Jacobson, S. L., Yindee, M., & Joshua, P. M. (2019, June 24-26). Methods for collecting ethnological data on Asian elephants with applications for conservation [Poster presentation]. In *The 2nd Young Conservation Scientists Conference*, Nakhon Ratchasima, Thailand
- Leelahapongsathon, K., **Horpiencharoen, W**., Thongyuan, S., Kasemsuwan, S., & Pinyopummintr, T. (2019). Rapid community dog assessment in rabies endemic area. *Proceedings of International Journal of Infectious Diseases*, Volume 79, 35. https://doi.org/10.1016/j.ijid.2018.11.098
- **Horpiencharoen, W**., Thongratsakul S. & Poolkhet C. (2017). *Risk factors on the presence of Streptococcus spp. in dairy cows with clinical mastitis in western Thailand* [Poster presentation]. The 14th National Kasetsart University Kamphaeng Saen Conference, Nakhon Pathom, Thailand

Skills and interests

Veterinary epidemiology | Infectious disease modelling | Wildlife conservation | Biodiversity | Zoonotic disease | Transboundary Animal Diseases |Public health | Geographic Information System

Proficiency in languages

Thai: Native

English: Fluent